Example 4.4

Find the expected value of the number of times a newborn baby's crying wakes its mother after midnight. The expected value is the expected number of times per week a newborn baby's crying wakes its mother after midnight. Calculate the standard deviation of the variable as well.

x	$P(x)$	$x*P(x)$	$(x-\mu)^2 \cdot P(x)$
0	$P(x=0)=\frac{2}{50}$	$(0)\left(\frac{2}{50}\right)=0$	$(0-2.1)^2 \cdot 0.04 = 0.1764$
1	$P(x=1)=\left(\frac{11}{50}\right)$	$(1)\left(\frac{11}{50}\right)=\frac{11}{50}$	$(1-2.1)^2 \cdot 0.22 = 0.2662$
2	$P(x=2)=\frac{23}{50}$	$(2)\left(\frac{23}{50}\right)=\frac{46}{50}$	$(2-2.1)^2 \cdot 0.46 = 0.0046$
3	$P(x=3)=\frac{9}{50}$	$(3)\left(\frac{9}{50}\right)=\frac{27}{50}$	$(3-2.1)^2 \cdot 0.18 = 0.1458$
4	$P(x=4)=\frac{4}{50}$	$(4)\left(\frac{4}{50}\right)=\frac{16}{50}$	$(4-2.1)^2 \cdot 0.08 = 0.2888$
5	$P(x=5)=\frac{1}{50}$	$(5)\left(\frac{1}{50}\right)=\frac{5}{50}$	$(5-2.1)^2 \cdot 0.02 = 0.1682$

Table 4.6 You expect a newborn to wake its mother after midnight 2.1 times per week, on the average.

Add the values in the third column of the table to find the expected value of X:

μ = Expected Value = $\frac{105}{50}$ = 2.1

Use μ to complete the table. The fourth column of this table will provide the values you need to calculate the standard deviation. For each value x, multiply the square of its deviation by its probability. (Each deviation has the format $x - \mu$).

Add the values in the fourth column of the table:

0.1764 + 0.2662 + 0.0046 + 0.1458 + 0.2888 + 0.1682 = 1.05

The standard deviation of X is the square root of this sum: $\sigma = \sqrt{1.05} \approx 1.0247$

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